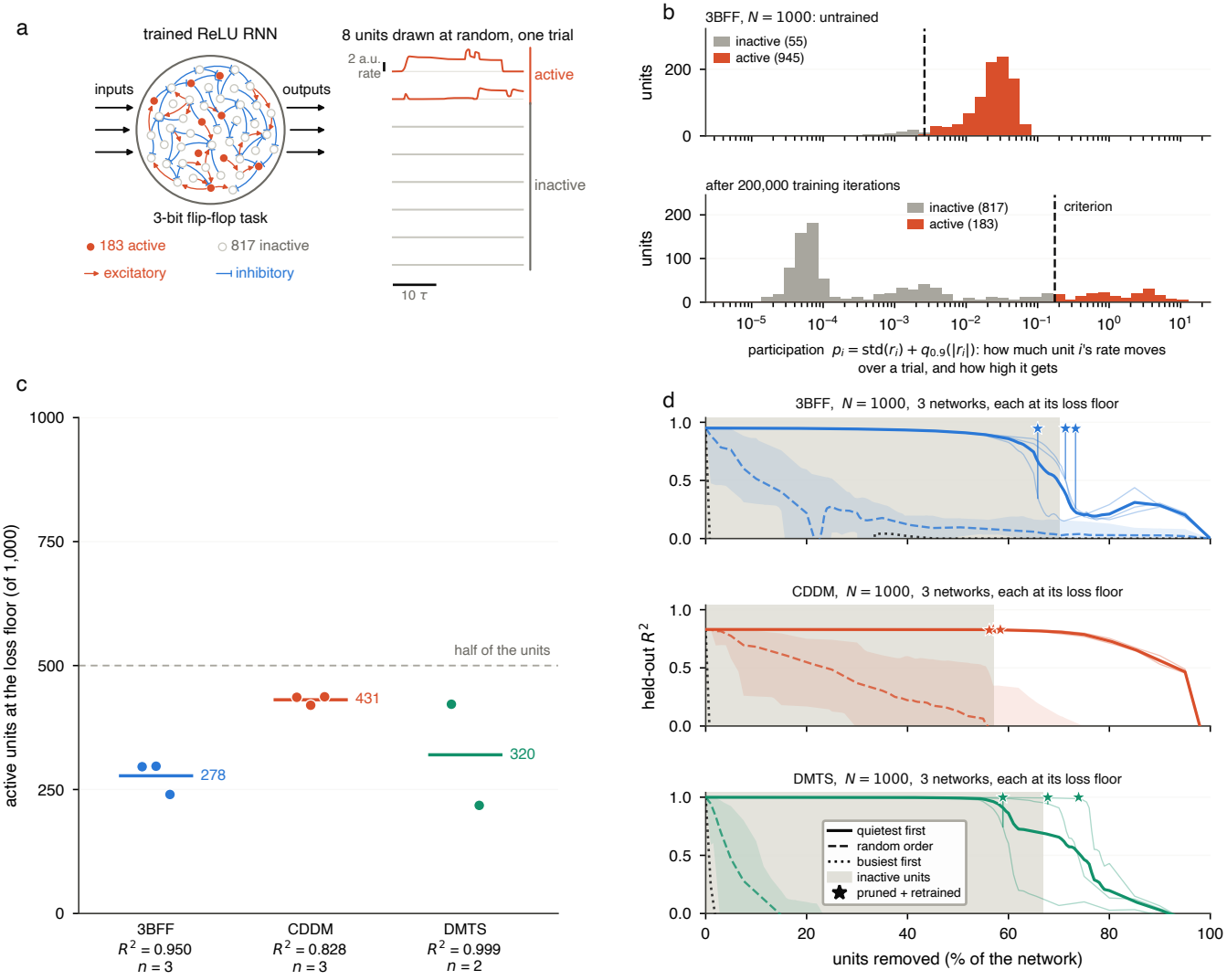

a
trained ReLU RNN
inputs
outputs
3-bit flip-flop task
183 active
817 inactive
excitatory
inhibitory
8 units drawn at random, one trial
2 a.u.
rate
active
inactive
10 τ
b
3BFF, N = 1000: untrained
inactive (55)
active (945)
units
after 200,000 training iterations
inactive (817)
active (183)
criterion
participation $p_i = \mathrm{std}(r_i) + q_{0.9}(|r_i|)$: how much unit i's rate moves over a trial, and how high it gets
c
active units at the loss floor (of 1,000)
half of the units
278
431
320
3BFF
R^2 = 0.950
n = 3
CDDM
R^2 = 0.828
n = 3
DMTS
R^2 = 0.999
n = 2
d
3BFF, N = 1000, 3 networks, each at its loss floor
CDDM, N = 1000, 3 networks, each at its loss floor
DMTS, N = 1000, 3 networks, each at its loss floor
held-out R^2
quietest first
random order
busiest first
inactive units
pruned + retrained
units removed (% of the network)